

# Unforgiving Confucian culture: A breeding ground for high academic achievement, test anxiety and self-doubt?

Lazar Stankov

Honorary Professor, School of Psychology, The University of Sydney, Sydney, NSW, 2006, Australia

## ARTICLE INFO

### Article history:

Received 31 December 2009

Received in revised form 26 April 2010

Accepted 11 May 2010

### Keywords:

Confucian culture  
Academic achievement  
Self-doubt  
Anxiety  
Unforgiving

## ABSTRACT

This paper reviews findings from several studies that contribute to our understanding of cross-cultural differences in academic achievement, anxiety and self-doubt. The focus is on comparisons between Confucian Asian and European regions. Recent studies indicate that high academic achievement of students from Confucian Asian countries is accompanied by higher levels of anxiety and self-doubt. After examining method, measurement, and context explanations of these findings, I argue that the culture of people living in contemporary Confucian Asian countries might be the driving force behind the combination of high achievement and negative psychological outcomes. Although forgiveness is a part of Confucian philosophy, people from modern Confucian Asian countries appear to be less forgiving than Europeans – i.e., they tend to disagree with statements that express toughness, maliciousness, and proviolence less strongly than Europeans. This relatively unforgiving attitude, coupled with the belief that effort rather than ability is the primary source of success, may be able to explain both high achievement and high anxiety and self-doubt among Confucian Asian students.

© 2010 Elsevier Inc. All rights reserved.

The aim of this paper is to bring together three related but hitherto somewhat separate areas of cross-cultural research, namely, studies of: a) educational achievement in cognitive domains; b) students' background variables that are non-cognitive in nature but have been found to be educationally important; and c) a broad range of non-cognitive and personality measures that are not part of a typical educational arsenal but are known to differ across cultures. International comparisons of academic achievement have been conducted for several decades now: the main findings from recent international assessments will be summarized briefly in the following section. Studies that link academic achievement to background variables are also not new, but in recent years there has been increased interest in this area of research. Finally, recent cross-cultural studies of personality, social attitudes, values and social norms seem to contain plausible clues as to the reasons for high anxiety and self-doubt in some high-achieving countries.

The focus of this paper will be on comparisons between Confucian Asian and European countries. House, Hanges, Javidan, Dorfman and Gupta (2004) provide justification for classifying all countries in the world into ten "societal clusters" or world regions on the basis of language, religion, geography, ethnicity and work-related values and attitudes. House et al. (2004) classification was based in part on Inglehart's (1997) use of World Value Survey data. In that work, a culture is defined with respect to the empirical evidence showing that a

group of countries have a common worldview. Inglehart (1997) employed factor analysis to determine the dimensionality of the space and a clustering procedure to group countries into world regions. All European countries will be referred to as European even though both House et al (2004) and Inglehart (1997) provide further subdivision of this continent. For example, one of the groupings of countries by Inglehart (1997) is labeled as Catholic Europe. It is important to keep in mind that Confucian Asian countries represent a world region with unique cultural features relative to other societal clusters. Confucian teachings are a part of this culture in the same way as Catholicism is a part of Latin American societal cluster (see Inglehart & Carballo, 1997). In other words, as used in this paper the label Confucian Asian refers to a geographic region with common cultural ties (i.e., a common worldview) that is distinct from, say, the Southern Asian societal cluster. In fact, I point out later on in this paper that perhaps it is possible to argue that recent historical and economic pressures in the region have led to the emergence of beliefs and behavior that are opposed to the basic tenets of Confucian philosophy.

The main purpose of this paper is to seek an improved understanding of cognitive and non-cognitive variables that impact achievement among secondary school students in Confucian Asian countries. Recent cross-cultural studies of personality, social attitudes, values and social norms (see Stankov & Lee, 2008, 2009) and attempts to develop scales of militant extremist mindset (Stankov, Saucier & Knežević, 2010) show Confucian Asians' stronger endorsement of statements that express toughness, maliciousness, and proviolence. These (anti-)social attitudes may be related to both high achievement and students' well-being that is evident from the large-scale studies of educational attainment. Is

E-mail address: [lazondi@rocketmail.com](mailto:lazondi@rocketmail.com).

students' pronounced anxiety and self-doubt too high price to pay for high achievement? If so, the clues for possible remedy may be within the culture itself.

## 1. Confucian Asians and Europeans: High achievement in Mathematics and Science

There has been an unprecedented surge of interest in the findings of recent international assessments such as the Trends in International Mathematics and Science Study (TIMSS) and the Programme for International Student Assessment (PISA). This increased interest could be attributed to a number of factors such as: the increased number of participant countries (e.g., 41 countries in PISA 2003), advanced techniques in large-scale assessments (e.g., spiral booklet design or computerized testing), ongoing year-to-year (trend) comparisons in the rankings of country performance, and increased public awareness via media and Internet.

The emphasis in TIMSS and PISA has been on student performance in mathematics, science, and reading literacy. PISA rotates its focus of main cognitive areas: in 2000 it was reading, in 2003 mathematics, and in 2006 science. According to Lee (2009) the top ten countries in terms of PISA 2003 mathematics scores easily divide into three groups: a) Four Confucian Asian countries (Hong Kong-China, South Korea, Japan and Macau-China); b) Five West European countries (Finland, The Netherlands, Liechtenstein, Belgium and Switzerland) and c) Canada. If we disregard the only North American country, we are left with two main groupings within the ten highest achieving countries – Confucian Asian and European countries. In terms of mathematics performance as it was assessed by TIMSS 1995 (Wilkins, 2004), the same two world regions top the list of the ten most successful countries: a) Confucian Asian (Singapore, Japan, South Korea, and Hong Kong); and b) East and West Europe (Belgium [Flemish], Czech Republic, The Netherlands, Slovak Republic, Switzerland, and Austria). Overall, in large-scale studies of educational attainment, Confucians tend to score somewhat higher than Europeans. This outcome is not restricted to mathematics; similar findings have been reported for science performance (Wilkins, 2004). More recent international studies (i.e., PISA 2006 and TIMSS 2007) are also in general agreement with this grouping of high-achieving countries. For example, the ten highest scoring countries in PISA 2006 assessment in Science are Finland, Canada, Japan, New Zealand, Hong Kong, Chinese Taipei, Estonia, Australia, The Netherlands and Korea. Similarly, TIMSS 2007 results for 8th graders' Mathematics scores provide the following order of the participating countries: Chinese Taipei, Korea, Singapore, Hong Kong, Japan, Hungary, England, Russian Federation, the United States and Lithuania.

### 1.1. Pedagogical and psychological issues that arose from the findings of Confucian–European comparisons

The above findings led to a lively debate in educational circles in the East Asian region and beyond. However, there was no comparable debate among psychologists.

#### 1.1.1. East Asia learner paradox?

Traditionally, educationists interested in cross-country comparisons have focused on East–West differences and perhaps not so much on more refined regional differences such as Confucian Asia versus Continental Europe. What became known as the East Asian Learner paradox (see Huang & Leung, 2005; Mok, 2006; Watkins & Biggs, 2001) is that some Westerners are puzzled by the fact that East Asian students do so well in international assessments despite having pedagogical practices that emphasize strategies of memorization and drill. This is in part because Western pedagogy tends to regard teacher-oriented classrooms as less conducive to deep learning. Watkins and Biggs (2001) argue that students from Confucian heritage countries

**Table 1**

PISA 2003 mean mathematics achievement scores and self-concept, self-efficacy and anxiety scores for Confucian Asian and European countries.

|               | Europeans                                            | Confucian Asians                     | Difference in Cohen's <i>d</i> |
|---------------|------------------------------------------------------|--------------------------------------|--------------------------------|
|               | Finland, Holland, Belgium, Switzerland, Lichtenstein | Hong Kong, Korea, Japan, Macau-China |                                |
| Math scores   | 531                                                  | 538                                  | 0.07                           |
| Self-concept  | −0.03                                                | −0.41                                | 0.38                           |
| Self-efficacy | 0.21                                                 | −0.31                                | 0.52                           |
| Anxiety       | −0.40                                                | 0.10                                 | 0.50                           |

\*For the achievement in math, mean = 500; standard deviation = 100. Self-concept, self-efficacy and anxiety scores are all in standard score units (e.g., mean = 0; standard deviation = 1).

are, in fact, deep learners despite the emphasis on repetitive effort. The existence of a paradox has also been challenged. For example, Hogan (2008) argues that no one has done the kind of cognitive and educational research that would allow the hypothesis to be tested properly. He also points out that East Asian pedagogy is well attuned to the aims of tasks set by TIMSS (i.e., the achieved curriculum, or what has been learned in school) and to a lesser extent to tasks that are more common in PISA (i.e., tasks that are designed to assess mathematical literacy, or what you can do in real life after you leave school)<sup>1</sup>. In general, the country differences on TIMSS assessments tend to be somewhat larger than on PISA assessments. The difference in the PISA 2003 mathematics scores between the five top-scoring European countries (mean = 531) and four Confucian Asian countries (mean = 538) is only 0.07 of the standard deviation (see first row in Table 1). This is not a substantial difference.

An unfortunate outcome of the debate regarding the East Asian Learner's Paradox may be reluctance to change a system that has consistently produced excellent results on international assessments of educational achievement i.e., an emphasis on repetition and memorization in schools in Confucian Asia. This could manifest in resistance to suggestions that it may be useful to introduce pedagogical practices emphasizing problem solving and critical and creative thinking. These skills are also covered under the broad umbrella term "21st Century skills", and it has been argued that these skills are critical in an increasingly complex and technologically sophisticated world.

Training students in thinking and problem solving is not new. Stankov (1986) described the dramatic effects achieved in the 1960s and 1970s in former Yugoslavia. In that part of Europe, just like in Confucian Asia today, the emphasis was on memorization. A series of experiments was carried out in two secondary schools, both of which shared the same facilities. Each experiment started with an incoming cohort of secondary school students (year nine, age 15). All students in both schools were administered a battery of cognitive measures of intelligence. The experimental group was then exposed to intensive (usually two to three times per week) classes involving training in creative or critical thinking and problem solving. Teachers in content areas were also taught how to incorporate principles of creative and critical thinking in their lessons. This same cohort was exposed to that type of instruction over a period of three years. Stankov (1986) reports gains in cognitive test scores in the experimental group compared with the control group after three years of training. The effects were an increase in scores on tests of intelligence, particularly fluid intelligence, that were 5 to 8 IQ points above those in the control group. This is important because, debates about intelligence aside (see Horn, 2008), tests of fluid intelligence measure problem solving skills and abilities that involve novel tasks and are independent of prior knowledge. Thus, training in critical and creative thinking and problem solving can produce superior outcomes in comparison to more

<sup>1</sup> Comparisons between TIMSS, PISA (and NAEP) assessments can be found on the following site: [http://nces.ed.gov/timss/pdf/naep\\_timss\\_pisa\\_comp.pdf](http://nces.ed.gov/timss/pdf/naep_timss_pisa_comp.pdf).

traditional pedagogical practices such as learning by rote. These results suggest that it would be worthwhile to study the relationship of TIMSS and PISA tasks to measures of intelligence. Are PISA scores more closely related to measures of intelligence than TIMSS scores? After all, measures of intelligence are known to predict many criteria of success in life. This might help to settle some long-running debates in this area (see below).

### 1.1.2. *Innate differences in ability?*

Some psychologists interested in individual and group differences in cognitive abilities attribute findings on PISA and TIMSS performance to (mostly innate) differences in ability. For example, [Rindermann \(2007\)](#) claims that IQ estimates are highly correlated with scores on PISA, TIMSS and PIRLS. [Lynn and Vanhanen \(2002, 2006\)](#) provide estimates of the average IQs for 81 countries. The IQ ranking is headed by five Confucian Asian countries (Hong Kong, South Korea, Japan, Taiwan, and Singapore); their average estimated IQ is 105. Five European countries (Austria, Germany, Italy, The Netherlands and Sweden) followed this Confucian Asian group, with an average estimated IQ of 102. This finding is roughly in agreement with the findings of ethnic differences within the United States: the average IQ of Asian Americans is said to be 103 whereas the IQ of Americans in all ethnic groups is 100<sup>2</sup>. It is sometimes pointed out that people of Confucian Asian origin studying in the US excel in math but not in abilities related to language (i.e., crystallized intelligence, Gc)<sup>3</sup>.

However, the overall ranking of countries with respect to IQ is still open to question. For example, a recent paper by [Brouwers, Van de Vijver and Van Hemert \(2009\)](#) reports the findings with Raven's Progressive Matrices test, arguably the best measure of fluid intelligence. In their data, Raven's Test places Norway on top of the list with mean raw IQ score of 132. Confucian Asian countries showed the range of IQ scores between 93 (Japan) and 109 (South Korea). Thus, at the very least, Raven's data provide a different ranking of countries than the IQ estimates of [Lynn and Vanhanen \(2002, 2006\)](#).

As is often the case in debates about intelligence, one of the problems is definitional — is intelligence a unitary trait or is it to be understood as multiple intelligences? Among those who subscribe to the latter conception, some would claim that mathematics test items in PISA and TIMSS assess both fluid (Gf), and crystallized (Gc) intelligence. If pure Gc tasks were to be used for estimating IQ, the ordering of countries would be different. For example, reading literacy is close to a Gc ability, which would result in ordering of countries that is different from orderings based on a math or science assessment<sup>4</sup>. Supporting arguments can be found in country performance on

reading achievement in the Progress in International Reading Literacy 2006 study (PIRLS; [Mullis, Martin, Kennedy, & Foy, 2007](#)). In this reading assessment, Russia and Hong Kong were listed as top performing countries together with other European countries that did not perform well in TIMSS 1995 and PISA 2003 assessments (e.g., Italy or Bulgaria)<sup>5</sup>. Again, the top five countries on the reading assessment in PISA 2000 were Finland, Canada, New Zealand, Australia, and Ireland ([Stanat et al., 2002](#)) with no Confucian Asian countries dominating the top list. Evidently, reading literacy performance does not uniformly place Confucian Asian countries on top of the rankings. Again, the claim that better academic performance of Confucian Asians is due to their higher innate ability is questionable when considering results based on reading achievement.

## 2. Confucian Asians versus Europeans: Differences on non-cognitive measures

Educational psychologists have provided an alternative explanation: that the higher achievement of Confucian Asian learners can be attributed to the effort they invest in studying (see [Huang & Leung, 2005](#); [Mok, 2006](#); [Watkins & Biggs, 2001](#)). The exertion of effort is related to motivation and, indeed, there is emerging evidence of significant differences among Confucian Asian and European countries on measures of motivation and other non-cognitive constructs. Of particular interest are those non-cognitive traits that are known to be both affected by cultural differences and are correlated with achievement scores. These may enhance or impede cognitive processes that are captured by large-scale international studies of educational achievement.

Despite the solid case pointing to a great success of Confucian Asian education systems, there are questions concerning non-cognitive variables that may also be important for success in school and perhaps even more so for psychological well-being. In both [Wilkins \(2004\)](#) and [Lee's \(2009\)](#) studies, pronounced differences between Confucian Asian and European countries were found on self-reported measures of anxiety (e.g., "I get very nervous doing mathematics problems"), self-efficacy (or confidence; e.g., "How confident do you feel about... being able to calculate how much cheaper a TV would be after 30% discount?") and self-concept (e.g., "I have always believed that mathematics is one of my best subjects"). Low levels on both self-concept and self-efficacy will be labeled self-doubt in the remainder of this paper in order to align their interpretation with anxiety. Overall, Confucian Asians are more anxious and self-doubting (i.e., they are less confident and have a lower self-concept scores) than Europeans ([Wilkins, 2004](#); [Lee, 2009](#)). [Wilkins \(2004\)](#) noted that on the measure of self-concept, European students are on top and Confucian Asian students are at the bottom among 41 countries. I calculated the means on self-concept, self-efficacy and anxiety scales based on [Lee's study \(2009\)](#) for five European and four Confucian Asian countries which performed well on PISA 2003 mathematics (see [Table 1](#)). The differences in Cohen's *d* (i.e., the effect sizes) between these two regions are 0.38 on self-concept, 0.52 on self-efficacy, and 0.50 on anxiety. The differences on self-efficacy and anxiety are five- to seven-fold larger than the difference on achievement between these two regions. The magnitude of the differences on non-cognitive measures is quite pronounced especially given that their cognitive performances do not show substantial differences.

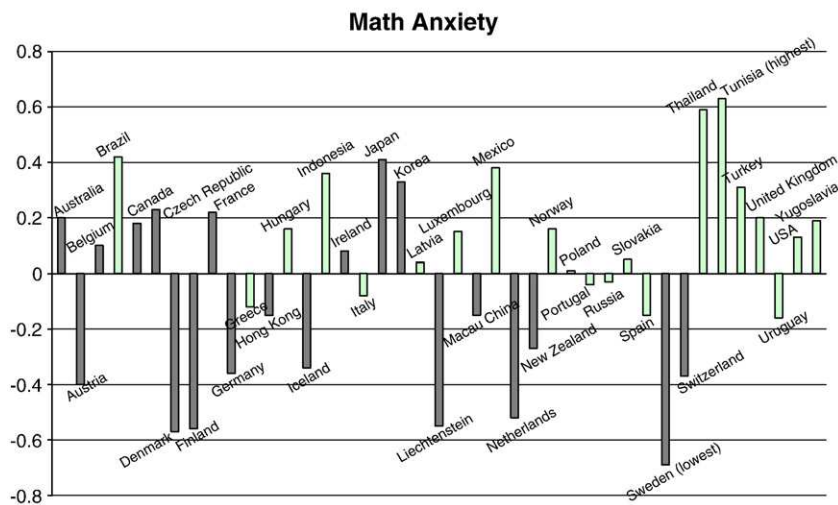
[Fig. 1](#) (from [Lee, 2009](#)) presents country means on the math anxiety scale from PISA 2003. We can examine the standing of four Confucian Asian countries (Hong Kong-China, South Korea, Japan and

<sup>2</sup> The latest available SAT 2009 results show that students of Asian origin obtained considerably higher scores than Whites on mathematics component of SAT and they were slightly worse on Critical Reading and about equal to Whites on Writing. It was reported that exam preparation experts are making claims that we need to find out what makes Asians perform this well and adopt their approach to coaching. The present paper cautions against uncritical acceptance of Asian approach on the grounds that the damage to students' well-being may be too high price to pay for higher achievement.

<sup>3</sup> In her PhD thesis at the University of Adelaide [Dandy \(2000\)](#) examined academic achievement and IQ of Australian students from Chinese and Vietnamese backgrounds. Her findings are quite similar to what is reported in the present paper. Thus, she concludes that "...it appears that Chinese and Vietnamese families place a strong emphasis on the value of education, and have high educational aspirations for their children, who in turn invest considerable effort in their studies. These values and behaviors combine to enhance academic achievements of Chinese and Vietnamese students."

<sup>4</sup> There are various components of reading literacy, but some (e.g. reasoning) might be more Gf — provided the educational system has equipped the student with the skills to read the sentences and draw the inferences. PIRLS looks at reading for literacy and informational purposes, and also straightforward inferences vs. higher-order thinking. Countries differ in rankings according to this e.g. Singapore, Hong Kong, and Taiwan are all significantly better at reading for informational rather than literacy purposes. Hong Kong is the only Asian country better at interpreting, evaluating, and integrating.

<sup>5</sup> Russia made major educational reforms between 2001 and 2006 that focused on improving reading literacy — see presentations at [http://www.iea.nl/49th\\_ga.html](http://www.iea.nl/49th_ga.html). There was also some controversy about the Russian results because Russian students were older than students in other countries.



**Fig. 1.** Math anxiety factor score means for all countries that participated in the Programme for International Student Assessment (PISA) in 2003. The dark shaded bars indicate countries whose math achievement scores are above average across all countries (obtained with permission, [Lee, 2009](#)).

Macau-China) and five West European countries (Finland, The Netherlands, Liechtenstein, Belgium and Switzerland) all of which were within the top 10 on PISA 2003 Mathematics performance. Confucian Asian countries, especially South Korea and Japan, show higher mean anxiety scores than European countries (with the exception of Belgium). The differences in self-doubt between Confucian Asian and European countries are equally or even more pronounced (see [Lee, 2009](#)). [Wilkins' \(2004\)](#) report based on TIMSS 1995 also showed that the lowest scoring countries on both math and science self-concept were Japan, South Korea and Hong Kong-China. Thus it appears that high self-doubt is a consistent finding in Confucian countries.

Also important to note is that the reported correlations in the PISA 2003 data between the measures of self-doubt and anxiety range from 0.45 to 0.67 in absolute values ([Lee, 2009](#)). These correlations are high enough to suggest the existence of a general factor underlying anxiety and self-doubting. Previous studies focusing on self-concept (e.g., [Wilkins, 2004](#)) have claimed that the low scores of Confucians' self-evaluation can be interpreted as a reflection of modesty (virtue) in Confucian Asian societies. However, significant correlations between self-doubt and anxiety signal the possible existence of a common source feeding both of these negative constructs because modesty is not generally considered to be a component of anxiety.

### 2.1. Is there a reason for concern?

What would be the reasons for low endorsements of self-concept and self-efficacy coupled with high anxiety among Confucian Asian students, despite their high academic performance? Four interpretations seem plausible. First, it could be that Confucian Asians, being more collectivist in orientation, tend to shun away from stating high levels on positive self-evaluation (e.g., self-concept and self-efficacy). If this argument holds true, there is little to worry about since results on self-reported non-cognitive measures may have to do more with self-presentation than "real" substantive differences (see [Heine & Hamamura, 2007](#)). One problem with this interpretation is that while self-presentation (i.e., modesty) may play some role in measures of confidence and self-concept, it is hard to interpret high levels of anxiety in the context of modesty even in Confucian Asian countries. Furthermore, the argument for self-presentation is typically derived from a comparison between Confucian Asians and North Americans, rather than between Confucian Asians and Europeans. There has been emerging evidence that Europeans, particularly East Europeans, and Asians share some similar psychological characteristics. For example,

it is reported that East Europeans stand in-between East Asian and Anglophone countries on both collectivism and holistic way of thinking (see [Nisbett & Miyamoto, 2006](#)). The extent to which these psychological characteristics influence self-presentation in the top ten PISA (2003) and TIMSS (2007) countries is not known.

The second interpretation is concerned with a measurement issue, pointing to less than perfect instruments being employed in large-scale assessments. Large-scale studies tend to focus on breadth (i.e., coverage of many constructs) and measure constructs with a small number of items. For example, the data reported in [Wilkins \(2004\)](#) were based on only two statements ("I usually do well in mathematics" and "I usually do well in science") to assess mathematics and science self-concepts respectively. A single item for the assessment of a construct is notorious for its lack of adequate reliability. Nevertheless, in TIMSS 2007 confidence was measured with 4 items, and Confucian Asian countries again showed up at the bottom of the list on math self-concept ([Mullis, Martin, & Foy, 2008](#)).

The third interpretation is that Confucian Asian students demonstrate a realistic representation of their confidence or self-related judgement within their own countries (i.e., within the context of high achievement, high standards and motivation). It is known that higher-performing students express higher confidence than their lower-achieving peers at a within-country level ([Lee, 2009](#)). Thus, a relatively larger portion of high performing students in Confucian Asian countries would necessarily entail lower ratings of themselves because they would judge themselves against their peers within their own country. Their self-view based on their peers in their own country would be a misrepresentation when it is compared to self-view of their peers in other countries. This is not a strong argument, of course, since a similar situation must exist in European countries as well. Differences in self-concept may also be related to differences in motivational belief. That is, perhaps Confucian Asian students see math to be less challenging than their western counterparts who "expect" math to be difficult ([Eaton, & Dembo, 1997; Leung, 2001, 2002](#)). This difference in motivational beliefs can play a part in Asian students' harsh judgements. For many years now it has been emphasized that societal forces, including family dynamics, exert a strong pressure to invest a lot of effort in academic success in Confucian Asian cultures — much more so than in most other cultural groups.

The fourth interpretation resides in the argument that the cross-cultural differences we observe are genuine, meaning that the three interpretations listed above (i.e., method effects, measurement issues, and context effect) are not strong enough to distort or reverse the phenomenon underlying "real" differences in psychological processes. The findings in recent PISA and TIMSS studies of a larger number of

young people in Confucian Asian societies who reported high levels of math anxiety and self-doubt may indicate the underlying pressure for academic achievement in these countries.

## 2.2. Are non-cognitive variables important?

It is important to ask whether the non-cognitive variables that we explore in this paper really matter in education and in a broader societal context. One may argue that high anxiety and self-doubt is a small price to pay for high achievement. Under this argument, a recommendation for those countries that do not demonstrate high levels of academic performance would be to motivate their students to work hard. On the other hand, there are countries that show negative outcomes in both areas: low achievement and high mean anxiety, like Tunisia or Thailand (see Fig. 1; Lee, 2009). What can be done under these conditions? Is anxiety acting in a way to shape the Yerkes–Dodson inverted U curve at the country level such that it is relatively low for European and relatively high in Confucian Asian regions? In other words, optimum level of arousal (anxiety) for high performance may be lower for Europeans than for Confucian Asians. Or perhaps it can be argued that Confucian Asians can tolerate higher anxiety without a detrimental effect on performance – i.e. they are more resilient (“tougher”).

### 2.2.1. Non-cognitive constructs and mental health

The findings reviewed up to now suggest a higher prevalence of negative feelings among Confucian Asian students compared to European students. This may be related to rapid industrialisation, relative lack of social safety nets such as cultural emphasis on looking after parents in old age as well as high social, employment, and marriage value of educational qualifications. It is generally accepted that Confucian Asian students experience tremendous stress due to familial and societal demands for academic success (Woo et al., 2004). However, this topic has been understudied so far and reliable epidemiological data are only now starting to emerge. Asia-based cross-cultural researchers point out that Confucian Asian students manifest their problems more internally than externally. Recent epidemiological studies of children's depression in Singapore indicate that reported rates of internalizing problems (withdrawal, somatic complaints and anxiety/depression) tend to be higher than the rates of externalizing (delinquent and aggressive behavior) problems in Singapore while the opposite is often observed in the West (Woo et al., 2007). Furthermore, it has been noted that psychological scales developed in the Western context may not be optimal scales for use in Confucian Asia. For instance, two components of a depression scale identified in the Confucian Asian participants are not typically found as salient aspects of depression in Western culture (Woo et al., 2004). One component is labeled as “socially oriented self-evaluation”, which has to do with concerns about social harmony. The other component labeled as “cognitive inefficiency” is related to concerns about school performance and career and it was the most salient worry for most Confucian Asian students. High levels of anxiety and self-doubt expressed among Confucian Asians in large-scale assessments can be related to this second component of depression, “cognitive inefficiency.” Literature on achievement motivation also stresses the fact that Asian students show keen interest in others' evaluations and reactions toward themselves and willingness to conform to the norms within their social networks (Bong, 2009; p. 281).

### 2.2.2. Non-cognitive constructs and academic achievement

There is abundant evidence of significant negative correlations between measures of students' non-cognitive constructs and their academic achievement (e.g., Alexander, Entwistle, & Dauber, 1993; Fincham, Hokoda, & Sanders, 1989; Lee, 2009; Voelkl, 1997). For instance, Lee (2009) reports moderate sizes of correlations between PISA 2003 math performance and math anxiety, math self-concept, and

math self-efficacy at both individual and country levels. The average correlations across all PISA 2003 participant countries at the individual level are reported to be  $-0.39$  with anxiety,  $0.43$  with self-efficacy/confidence, and  $0.23$  with self-concept (Lee, 2009). Other researchers have reported similar moderate-sized negative correlations of math performance and math anxiety (Ma, 1999; Hembree, 1990), and positive correlations with self-concept (Marsh, Hau, Artlet, Baumert, & Peschar, 2006) and self-efficacy (Scholz, Dona, Sud, & Scuarzer, 2002). At the country level of analysis, the correlations are also significant. The country level PISA 2003 mathematics performance showed the correlations of  $-0.65$  with anxiety and  $0.42$  with self-efficacy/confidence (Lee, 2009).

## 3. Beyond education: Unforgiving Confucian Asians and tolerant Europeans

Confucian Asian culture has a long history of high regard for learning and achievement and emphasis on effort to achieve academically. Its collectivist aspect underscores relationships, family closeness, and social harmony. Putting together these two salient features of Confucian Asian culture leads to the perception that individuals strive to achieve not only for their personal success but also for honor of their family and society (see Huang & Leung, 2005; Mok, 2006; Watkins & Biggs, 2001). A finding from PISA 2003 that Confucian Asian students expressed higher levels of anxiety and self-doubt (Lee, 2009) can be interpreted in terms of this unique cultural aspect of Confucianism. That is, in the minds of Confucian Asian students, the distinction between the self and one's family is not clear-cut and self achievement is also seen as family's achievement. Consequently, Confucian Asian students become aware of and learn to take seriously the implications and consequences of their academic success and failure. From this vantage point, the internal pressure for academic achievement is probably higher in Confucian Asian societies than in the other parts of the world.

### 3.1. Broader context of cross-cultural differences

Large-scale cross-cultural studies are not limited to the field of education. As a reaction to globalization, social and industrial/organizational psychologists have become interested in cross-cultural differences in non-cognitive constructs. Psychological domains that showed advances in our understanding of cross-cultural differences are found in the areas of personality (McCrae & Terracciano, 2008), social attitudes (Stankov & Knežević, 2005; Saucier, Akers, Miller, Stankov, & Knezevic, 2009), values (Schwartz, 2003) and social norms (House et al., 2004). Recently, Stankov and Lee (2008, 2009) put the measures of these four major domains into a single battery and administered it to over 2000 participants aged between 18 and 21 from 73 countries. The number of participants in some countries was small, so countries were combined into world regions or societal clusters according to GLOBE study (House et al., 2004). These world regions are indicated on the horizontal axis in Fig. 2.

Findings of Stankov and Lee (2008, 2009) show that while Confucian Asians did not differ from the rest of the world on Conservatism and Social Norms factors, they were lower than all other world regions on a factor consisting of personality and anti-social attitudes which include the scales of Toughness and Maliciousness (for more detailed results, see Stankov, 2007; Stankov & Lee, 2008, 2009). Fig. 2 presents arithmetic means of the combined Toughness and Maliciousness scales using the data from Stankov and Lee (2009) studies. The vertical axis in Fig. 2 ranges from 1 to 5: the items employed are based on a five-point Likert scale ranging from “strongly disagree” to “strongly agree”. The scales of Toughness and Maliciousness consisted of twenty items each. The development of the English versions of these scales was reported in Stankov and Knežević (2005). The items date back to early 1990s when a group of Serbian psychologists developed new scales of psychoticism

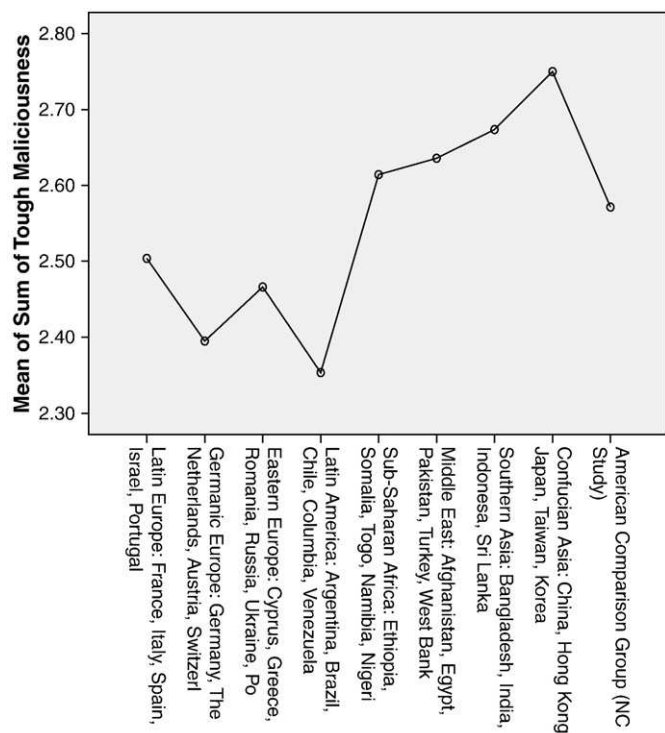


Fig. 2. Arithmetic means for nine world regions on a composite score of Toughness and Maliciousness scales (obtained with permission, Stankov and Lee, 2008; 2009).

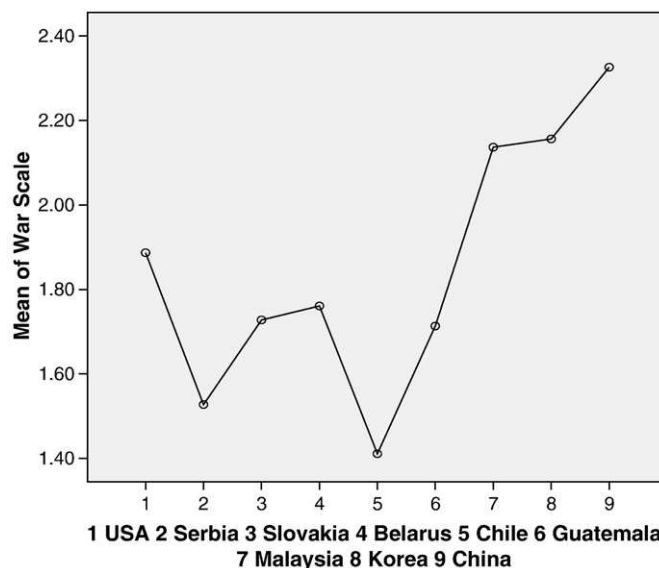


Fig. 3. Arithmetic means for nine countries on Provivence scale (obtained with permission, Stankov et al, 2009).

and related constructs. The Toughness scales indicate machismo and tough personality, including such items as: “I am tired of being forced to do things.”, “Rules can kill your will to live.” and “I want to do whatever pleases me, regardless of who is in my way”. The Maliciousness scale contains items that endorse anger, violence and brutality, such as “If I had complete power over people, many would regret the day they were born”, “The only difference between criminals and others is that criminals have the courage to act on their desires”, and “Revenge that is planned over a long period of time is always sweeter than quick revenge”. Two points can be made in regards to the plot in Fig. 2. First, Confucian Asian countries have higher average scores on the Toughness/Maliciousness measure than the other eight world regions. The differences between the means of the nine world regions in Fig. 2 are statistically significant  $F(8,1976) = 24.561$ . Second, European and Latin American world regions score lower than all other world regions on this Toughness/Maliciousness measure. In fact, the difference between Confucian Asia and Latin America is exactly one standard deviation. Germanic Europe scores closer to Latin America and the difference is only slightly lower than a full standard deviation. This is substantial. In addition, it is worth noting that Confucian Asians’ mean of 2.75 is the closest to 3—i.e., a neutral Likert Scale point—indicating that a larger number of people, but not a majority, in Confucian Asia endorse these negative attitudes more strongly than people from other parts of the globe.

A similar finding was observed in a study by Stankov, Saucier, and Knežević (2010). This study was designed to assess militant extremist mindset (MEM) underlying fanatical thinking of those involved in terrorist activities. One of the subscales of MEM is labeled as Provivence (i.e., pro-war attitudes). Example items include: “Killing is justified when it is an act of revenge”, “If violence does not solve problems, it is because there was not enough of it”, “Our enemy’s children are like scorpions; they need to be squashed before they grow up”, and “The only way to teach a lesson to our enemies is to threaten their lives and make them suffer”. This scale was given to first year university students from nine countries ( $N = 2424$ ). Again, Fig. 3 shows that Confucian Asian countries (Korea and China) had higher values on this scale than the other countries sampled in this

study. And again, European and Latin American countries scored lower than countries from all other world regions on the Provivence scale. The differences between the means for all nine countries were significant ( $F[8,2415] = 86.77, p < 0.01$ ).

Overall, participants endorse Provivence (Fig. 3) statements less strongly than Toughness/Maliciousness (Fig. 2) statements. It is important to note that the distributions for both measures are clearly skewed towards the lower end of “strongly disagree” and “disagree”. In other words, the comparisons between Europeans and Confucian Asians were made in relative rather than absolute terms since people in general did not approve of nastiness or violence. Nevertheless, the general conclusions related to Confucian Asians versus Europeans are identical in both Figs. 2 and 3: Confucian Asian countries (e.g., Korea and China) showed higher average scores on the Provivence measure than the other countries sampled in these studies. Also, European and Latin American countries scored lower than countries from all other world regions on the Toughness/Maliciousness and Provivence scales.

The Oxford English Dictionary defines forgiveness as “to grant free pardon and to give up all claim on account of an offense or debt”. Or, as expanded in the Wikipedia entry “Forgiveness is typically defined as the process of concluding resentment, indignation or anger as a result of a perceived offense, difference or mistake, and/or ceasing to demand punishment or restitution.”<sup>6</sup> I may add that tolerance in the sense of lenience and acceptance is close to the meaning of forgiveness in this paper. However, I prefer using the term unforgiveness since the antonym of tolerance (i.e., “intolerance”) implies bigotry, prejudice and fanaticism.

The interpretation of the data from both Figs. 2 and 3 is that people in Confucian Asian countries are relatively unforgiving and people in European and Latin American countries are relatively tolerant compared to the other parts of the world. This unforgiving attitude of Confucian Asian people, coupled with their belief that effort and hard work can lead to academic success (see Watkins & Biggs, 2001), is a likely cause for considerable stress experienced by Confucian Asian students. My argument is that high academic achievement and high levels of anxiety and self-doubt among Confucian Asian students, which were observed in recent TIMSS and PISA studies (Lee, 2009; Wilkins, 2004), may be due to an unforgiving element of contemporary culture

<sup>6</sup> Retrieved on April 25, 2010.

in Confucian Asian countries. We need properly designed empirical studies to examine what seems to be a plausible causal link further.

As pointed out by the unknown reviewers, when talking about beliefs and attitudes within collectivist cultures (e.g., Confucian Asians) it may be important to distinguish between in-group and out-group membership. Thus, collectivists tend to support and encourage (forgive?) more strongly than people from other world regions in-group members and exclude and reject (not forgive?) out-group members. Although this is an important point, the available data do not allow us to examine this issue further. The evidence to date indicates that, in general, Confucian societies tend to be less forgiving than societies from other parts of the world. It is reasonable to assume that this is likely to transfer to in-group members, especially if school achievement is highly valued. But empirical studies are needed in order to examine this issue.

It is noteworthy that a recent study which compares New Zealand and Chinese participants from Hong Kong on several direct measures of forgiveness agrees with the findings reported above. Hook et al. (in press) report that New Zealanders are more forgiving, have lower levels of revenge motivation and are more accepting of conciliatory gestures than are the Chinese. Relatedly, Neff, Pisitsungkagarn and Hsieh (2008) report cross-cultural comparisons of samples from the United States, Thailand, and Taiwan on measures of self-compassion and self-criticism. They found that Taiwanese (Confucian culture) were least compassionate among the three groups examined. Thus, the message is the same irrespective of whether one tries to measure unforgiveness (indirectly) or forgiveness (directly): Confucian Asians tend to endorse more strongly statements showing unforgiveness than people of European origin.

#### 4. Effort, guilt-driven anxiety and harsh punishment for failure

I wish to emphasize in closing that empathy and forgiveness are an integral part of most great religious systems, including Confucian philosophy. The source of unforgiveness cannot be traced back to Confucian teachings. Two readers of an earlier draft of this paper pointed out to the well-known quote from Confucius: When one of his students asked him, “Is there one word that guides us to do the right thing the whole life?”, he replied, “Forgiveness”. Rather, the unforgiving nature of Confucian Asian societies that is apparent in the available data should be interpreted as a psychological mindset resulting from a multitude of collective experiences among modern Confucian Asian peoples. In other words, crucial for the unforgiving mindset may be recent socio-political past, rapid economic development and population density with concomitant competition for the available resources within the region rather than Confucian philosophy itself. These historical developments might have exacerbated features of Confucian culture that make it appear unforgiving in comparison to other regions of the world<sup>7</sup>.

As alluded earlier, this recent history may feed on the particular importance that Confucian heritage attributes to effort. For example, Lee (1996) states that “... there is an extraordinary emphasis on effort, willpower or concentration of the mind in the Confucian tradition. Because there is a strong belief in attainability by all, there is also a strong belief that one's failure is not due to one's internal make-up or ability, but one's effort and willpower.” (p. 39) Within the collectivist

value system that emphasizes belongingness of an individual to the family and other social structures the emphasis on effort provides one explanation of the findings reviewed in this paper. Internally, this can lead to the highlighted feeling of guilt and anxiety in the face of failure. Externally, the emphasis on effort can lead to harsh and punishing judgement of others' failures<sup>8</sup>. Overall, Confucian Asians are more likely to attach harsher negative set of emotional attributions to both internal and external causes – i.e., they appear to be more critical about selves and others.

This analysis suggests that from among the three actors in the educational arena in Confucian countries—*student* characteristics, *teacher* expertise and practices and *parents'* attitudes towards children's education—it is the parents who are the most important. The role of parents is particularly emphasized by Kom and Park (2006) in their analyses of the academic achievement in Korea. Curiously, even though Stevenson and Stigler (1992) have pointed to the crucial role of parental attitudes in children's education, investment in educational research in Singapore and Hong Kong at least has focused on pedagogy as witnessed by the East Asian Learner's paradox debate mentioned earlier.

#### 5. Educational interventions versus societal changes

Recently, educational authorities in several Confucian Asian countries took steps to raise the awareness about the importance of non-cognitive aspects of school life. The outbreak of teen suicides in Japan in the early 2000s was attributed to both stress to achieve academically and the culture of bullying at school, which led to widespread discussions and steps to improve school climate in that country. In Singapore, the government decided to introduce school counselors to all schools in 2005. Since then, social/emotional learning has become one of the important tenets of Singaporean education.

Educational interventions carried out under the label of “social and emotional learning” have limitations. At best, these types of interventions help victims but can't really remove the causes. The causes more likely reside within the community, culture, and society as a whole. I suggest a two-pronged approach that may deal with the causes effectively. First, public awareness of the psychological suffering of young children due to high pressure for academic attainment (especially in comparison with children from other countries) can be increased through policy measures and media campaigns. Emphases can be placed on the importance of psychological well-being for young students and a more tolerant view of parents and society in general towards children's' academic accomplishment. Perhaps also a reduced emphasis on conformity and tolerance of social outliers may be helpful. Second and potentially more important is that the governments in Confucian Asian countries move towards improving the desirability (e.g., by increasing the income) of jobs that do not require high academic achievement so that students and parents would be aware of opportunities that may come later on in life even for those who did not perform well on exams. An increase in the number of alternative routes to higher education would also provide a safety net for those students who don't succeed in school to have a chance later on in life. Both these approaches seem to have worked elsewhere, most notably in Europe. European countries set an example that high academic achievement can be obtained with low levels of anxiety and self-doubting.

<sup>7</sup> I wish to call reader's attention to the fact that Southern Asian cluster in Fig. 2 and Malaysia in Fig. 3 score only slightly lower than Confucian countries. They are geographically adjacent to Confucian Asian cluster and have been experiencing similar rapid economic development along with turbulent recent history. These countries do not share Confucian heritage. We do not have PISA or TIMSS data for Southern Asian countries and therefore we cannot compare them in terms of achievement, anxiety and self-doubt. However, relatively high level of unforgiveness in these countries supports the thesis that Confucian philosophy on its own does not play a major role in generating toughness and maliciousness. Recent societal pressures appear to be a more reasonable explanation for such beliefs and behaviors.

<sup>8</sup> Since educational systems in Confucian Asian countries differ, with Singapore and Hong Kong being influenced by the British and Taiwan, Korea and Japan relying on their indigenous traditions in education, we may ask whether instructional practice and pedagogy make much difference. The effects of intervention and pedagogical innovation may be insignificant in comparison to the amount of effort students in Confucian Asian countries invest into studying. From this perspective, the debate about East Asian Learner Paradox that was mentioned earlier in this paper was focused in part on an irrelevant issue.

Ultimately, of course, it may be seen as useful to introduce measures that may mellow the unforgiving mindset of people living in Confucian Asian countries in a broader sense, not only in relationship to the young. This change may also be accomplished through a joint use of formal education, media and public policy measures and by targeting all aspects of unforgiveness. Comparisons to other world regions may provide a mirror that can be usefully employed in this societal re-education process.

## 6. Summary and conclusions

Confucian Asian students possess a strong drive towards achievement, which is deeply rooted in the Confucian culture. We have also observed that Confucian Asian countries demonstrate less forgiveness towards underachievement and misbehaviors. This psychological make-up certainly brings about the effort and persistence needed for high levels of academic performance. Yet, another piece in this story is that Confucian Asian students experience high anxiety and self-doubt compared to students in other parts of the world whose performance is only slightly lower. Despite their strong academic performance, the presence of higher mean anxiety/self-doubt should be of some concern to educators and to societies as a whole.

It is likely that economic development that took place in East Asia over the past half century owes something to the perception that educational attainment is one of the most effective ways to move out of poverty. Now that economic prosperity has been reached at least among the urban areas in several countries in the region, the time may be ripe to consider the costs and seek ways to redress the balance. In other words, people in Confucian Asian countries may want to consider the question “Is high achievement worth the suffering that seems to be associated with it?” If not, systematic effort both within the educational system and within the whole of society should be considered.

Caveats are in order, of course. It may be argued that the evidence that Confucian Asian countries have problems (and by implication Europe does not) is not strong enough. In fact, the evidence from PISA and TIMSS on achievement and anxiety/self-doubt has been replicated in large-scale studies. The evidence about the cross-cultural differences on measures of unforgiveness (i.e., social attitudes of Toughness/Maliciousness and Provivence) is new and weaker. However, this evidence is consistent across two cross-cultural studies.

My argument is simply that the issues discussed in this paper are worthy of further investigation. The most pressing is the need to establish an empirical link both at the individual and aggregate levels between measures of unforgiveness on one hand and achievement and anxiety/self-doubt on the other. In the absence of such evidence, much of what has been said in this paper will remain speculative and question mark cannot be removed from its title.

## References

- Alexander, K. L., Entwistle, D. R., & Dauber, S. L. (1993). First-grade classroom behavior: Its short- and long-term consequences for school performance. *Child Development*, 64, 801–814.
- Bong, M. (2009). Age-related differences in achievement goal differentiation. *Journal of Educational Psychology*, 101, 879–896.
- Brouwers, S. A., Van de Vijver, F. J. R., & Van Hemert, D. A. (2009). Variation in Raven's Progressive Matrices scores across time and place. *Learning and Individual Differences*, 19, 330–339.
- Dandy, J. (2000). IQ and academic achievement among Australian students from Chinese and Vietnamese backgrounds. PhD Thesis, University of Adelaide.
- Eaton, M. J., & Dembo, M. H. (1997). Differences in the motivational beliefs of Asian American and non-Asian students. *Journal of Educational Psychology*, 89, 433–440.
- Fincham, F. D., Hokoda, A., & Sanders, R. (1989). Learned helplessness, test anxiety, and academic achievement: A longitudinal analysis. *Child Development*, 60, 138–145.
- Heine, S. J., & Hamamura, T. (2007). In search of East Asian self-enhancement. *Personality and Social Psychology Review*, 11, 1–24.
- Hembree, R. (1990). The nature, effects, and relief of mathematics anxiety. *Journal for Research in Mathematics Education*, 21, 33–46.
- Hogan, D. (2008, June). Pedagogical practice and pedagogical research in South East Asia: The case of Singapore. Paper presented at the International Symposium of Teaching, Learning and Assessment, Hong Kong.
- Hook, J. N., Worthington, E. L., Davis, E. E., Watkins, D., Hui, E., Luo, W., Fu, H., Shulruf, B., Morris, P. (In press). A China–New Zealand comparison of forgiveness. *Journal of Cross-cultural Psychology*.
- Horn, J. L. (2008). Spearman, g, expertise and the nature of human cognitive capability. Ch. 10. In P. C. Kyllonen, R. D. Roberts, & L. Stankov (Eds.), *Extending intelligence: Enhancement and new constructs*. Mahwah, NJ: Lawrence Erlbaum Associates.
- House, R. J., Hanges, P. J., Javidan, M., Dorfman, P. W., & Gupta, V. (2004). *Culture, leadership, and organizations: The GLOBE study of 62 societies*. Thousand Oaks, CA: SAGE Publications.
- Huang, R., & Leung, F. K. S. (2005). Deconstructing teacher-centeredness and student-centeredness dichotomy: A case study of a Shanghai mathematics lesson. *The Mathematics Educator*, 15, 35–41.
- Inglehart, R. (1997). *Modernization and postmodernization: Cultural, economic and political change in 43 societies*. Princeton: Princeton University Press.
- Inglehart, R., & Carballo, M. (1997). Does Latin America exist? (And is there a Confucian culture?). A global analysis of cross-cultural differences. *Political Sciences and Politics*, 30(1), 34–46.
- Kom, U., & Park, Y. -S. (2006). Indigenous psychological analysis of academic achievement in Korea: The influence of self-efficacy, parents, and culture. *International Journal of Psychology*, 41, 287–292.
- Lee, W. O. (1996). The cultural context for Chinese learners: Conceptions of learning in the Confucian tradition. In D. A. Watkins, & B. Biggs (Eds.), *The Chinese Learner: Cultural, Psychological and Contextual Influences* (pp. 25–41). Comparative Education Research Centre and Australian Council for Educational Research: Hongkong and Melbourne.
- Lee, J. (2009). Universals and specifics of math self-concept, math self-efficacy, and math anxiety across 41 PISA 2003 participating countries. *Learning and Individual Differences*, 19, 355–365.
- Leung, F. K. S. (2001). In search of an East Asian identity in mathematics education. *Educational Studies in Mathematics*, 47, 35–51.
- Leung, F. K. S. (2002). Behind the high achievement of East Asian students. *Educational Research and Evaluation: An International Journal on Theory and Practice*, 8, 87–108.
- Lynn, R., & Vanhanen, T. (2002). *IQ and the wealth of nations*. Westport, CT: Praeger.
- Lynn, R., & Vanhanen, T. (2006). *IQ and global inequality*. Athens: Washington Summit.
- Ma, X. (1999). A meta-analysis of the relationship between anxiety toward mathematics and achievement in mathematics. *Journal for Research in Mathematics Education*, 30, 520–540.
- Marsh, H. W., Hau, K. -T., Artelt, C., Baumert, J., & Peschar, J. L. (2006). OECD's brief self-report measure of educational psychology's most useful affective constructs: Cross-cultural, psychometric comparisons across 25 countries. *International Journal of Testing*, 6, 311–360.
- McCrae, R. R., & Terracciano, A. (2008). The Five-Factor model and its correlates in individuals and cultures. Ch. 10. In F. J. R. Van de Vijver, D. A. Van Hemert, & Y. H. Poortinga (Eds.), *Multilevel analyses of individuals and cultures* (pp. 249–285). Mahwah, NJ: Erlbaum.
- Mullis, I. V. S., Martin, M. O., & Foy, P. (2008). *TIMSS 2007 international mathematics report: Findings from IEA's trends in international mathematics and science study at the fourth and eighth grades*. Chestnut Hill, MA: TIMSS & PIRLS International Study Center, Boston College with Olson, J. F., Preuschoff, C., Erberber, E., Arora, A., & Galia, J.
- Mullis, I. V. S., Martin, M. O., Kennedy, A. M., & Foy, P. (2007). *PIRLS 2006 international report: IEA's progress in international reading literacy study in primary schools in 40 countries*. Chestnut Hill, MA: Boston College.
- Mok, I. (2006). Shedding light on the East Asian learner paradox: Reconstructing student-centredness in a Shanghai classroom. *Asia Pacific Journal of Education*, 26, 131–142.
- Neff, K. D., Pisitsungkarn, K., & Hsieh, Y. (2008). Self-compassion and self-construal in the United States, Thailand, and Taiwan. *Journal of Cross-Cultural Psychology*, 39, 267–285.
- Nisbett, R. E., & Miyamoto, Y. (2006). The influence of culture: Holistic vs. analytic perception. *Trends in Cognitive Sciences*, 9, 467–473.
- Rindermann, H. (2007). The g-factor of international cognitive ability comparisons: The homogeneity of results in PISA, TIMSS, PIRLS and IQ-tests. *European Journal of Personality*, 21, 667–706.
- Saucier, G., Akers, L. G., Miller, S. S., Stankov, L., & Knezevic, G. (2009). Patterns of thinking in militant extremism. *Perspectives on Psychological Science*, 4, 256–271.
- Scholz, U., Dona, B. G., Sud, S., & Scuwazer, R. (2002). Is general self-efficacy a universal construct? Psychometric findings from 25 countries. *European Journal of Psychological Assessment*, 18, 242–251.
- Schwartz, S. H. (2003). Basic human values: Their content and structure across cultures. In A. Tamayo, & J. Porto (Eds.), *Valores e trabalho [Values and work]*. Brasília: Editora Universidade de Brasília.
- Stanat, P., Artelt, C., Baumert, J., Klieme, E., Neubrand, M., Prenzel, M., et al. (2002). *PISA 2000: Overview of the study, design, method and results*. Berlin: MaxPlanck-Institut.
- Stankov, L. (1986). Kvashchev's experiment: Can we boost intelligence? *Intelligence*, 10, 209–230.
- Stankov, L. (2007). The structure among measures of personality, social attitudes, values, and social norms. *Journal of Individual Differences*, 28, 240–251.
- Stankov, L., & Knežević, G. (2005). Amoral social attitudes and value orientations among Serbs and Australians. *Australian Journal of Psychology*, 57, 115–128.
- Stankov, L., & Lee, J. (2008). Culture: Ways of thinking and believing. In G. J. Boyle, G. Matthews, & D. Saklofske (Eds.), *The handbook of personality theory and testing* (pp. 560–575). : Sage Publications.
- Stankov, L., & Lee, J. (2009). Dimensions of cultural differences: Pancultural, etc/emic, and ecological approaches. *Learning and Individual Differences*, 19, 339–354.

- Stankov, L., Saucier, G., & Knežević, G. (2010). Militant extremist mindset: Provioleence, vile world, and divine power. *Psychological Assessment*, 22(1), 70–86.
- Stevenson, H. W., & Stigler, J. W. (1992). *The learning gap: Why our schools are failing and what we can learn from Japanese and Chinese education*. New York: A Touchstone Book published by Simon and Schuster.
- Voelkl, K. E. (1997). Identification with school. *American Journal of Education*, 105, 294–318.
- Watkins, D.A., and Biggs, J.B. (2001). The paradox of the Chinese learner and beyond. In D.A. Watkins and J.B. Biggs (Eds.), *The Chinese learner: Cultural, psychological, and contextual influences* (pp. 3–26). Hong Kong: Comparative Education Research Centre, The University of Hong Kong; Melbourne, Australia: Australian Council for Education Research.
- Wilkins, J. M. (2004). Mathematics and science self-concept: An international investigation. *The Journal of Experimental Education*, 72, 331–346.
- Woo, B. S. C., Chang, W. C., Fung, D. S. S., Koh, J. B. K., Leong, J. S. F., Kee, C. H. Y., et al. (2004). Development and validation of a depression scale for Asian adolescents. *Journal of Adolescence*, 27, 677–689.
- Woo, B. S. C., Ng, T. P., Fung, D. S. S., Chan, Y. H., Lee, Y. P., Koh, J. B. K., et al. (2007). Emotional and behavioural problems in Singaporean children based on parent, teacher and child reports. *Singapore Medical Journal*, 48, 1100–1106.